S181M116 Derivatives and Alternative Investments: Lecture 5

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This lecture develops the stochastic calculus used in continuous-time option pricing. We begin with stochastic processes and the Markov property, then build Brownian motion from independent normal increments. We add drift and volatility, specialize the model to stock prices through geometric Brownian motion, and use Monte Carlo simulation to interpret the model. The main mathematical tool is Ito's lemma, which explains how functions of stochastic variables evolve. The final result is the lognormal distribution of stock prices, which is the input for the Black-Scholes-Merton model in Lecture 6.

Reading. Hull, Chapter 14. The goal is not to turn students into probability theorists. The goal is to understand the stochastic notation well enough that the Black-Scholes-Merton derivation in the next lecture feels like economics plus no-arbitrage, not mysterious algebra.

Roadmap

Lecture 4 ended with binomial trees. A binomial tree is a discrete-time model: over each short period the stock either moves up or down. Lecture 5 asks what happens when the time step becomes very small and the stock is allowed to move continuously.

The lecture has five blocks:

1. **Stochastic processes and Markov dynamics.** Define the objects whose future is uncertain.
2. **Brownian motion.** Build the basic source of continuous-time randomness.
3. **Drift, volatility, and stock prices.** Explain why stock prices are modeled with percentage, not dollar, shocks.
4. **Correlated processes and simulation.** Connect the math to practical Monte Carlo paths.
5. **Ito's lemma and lognormality.** Learn the chain rule of stochastic calculus and derive the distribution of future stock prices.

Main lesson. In ordinary calculus, small changes are of order dt. In stochastic calculus, Brownian shocks are of order $\sqrt{dt}$. This is why the square of a random shock matters: $(dz)^2$

behaves like dt and creates the extra $\frac{1}{2}$ term in Ito's lemma.

Stochastic Processes

What Is a Stochastic Process?

A stochastic process is a collection of random variables indexed by time. If X_t is the process, then X_t is the value of the variable at time t.

Definition 1 (Stochastic process)
A stochastic process is a rule that assigns a random variable X_t to each date t in a time set. The time set may be discrete, such as $t = 0, 1, 2, \ldots$, or continuous, such as $t \in [0, T]$.

There are two separate distinctions.

Discrete time versus continuous time A process is observed or updated only at fixed dates in discrete time. In continuous time, changes can occur at any instant.

Discrete state versus continuous state A process has a discrete state space if it can take only certain values. It has a continuous state space if it can take any value in an interval.

A binomial stock tree is discrete time and discrete state. A continuously traded stock model is usually treated as continuous time and continuous state, even though real exchanges trade in ticks and only during market hours.

Model versus reality. Continuous-time models are idealizations. Stock prices do not literally change at every mathematical instant, and quotes are not infinitely divisible. The model is useful because it gives tractable approximations and sharp no-arbitrage restrictions.

The Markov Property

Definition 2 (Markov property)
A process has the Markov property if the conditional distribution of the future depends on the current state, but not on the path taken to reach that state. Formally, for $u > t$,

$$\mathbb{P}(X_u \leq x \mid X_t, X_{t_1}, \ldots, X_{t_n}) = \mathbb{P}(X_u \leq x \mid X_t),$$

for all past dates $t_1, \ldots, t_n < t$.

For a stock price, the Markov assumption says that if the stock is worth 100 now, then the distribution of its future price is summarized by the current price and current model parameters. The fact that it was 90 last month or 120 last month does not directly matter after we condition on today's price.

> **Economic intuition.** A Markov stock price is the continuous-time cousin of weak-form market efficiency. If past price patterns contained reliable profit opportunities, traders would act on them, and their trades would tend to move the information into today's price.

The Markov property does not say that past data are useless. Historical prices can help estimate parameters such as volatility. It says that, after the model parameters are known, the current state is enough for forecasting the future distribution.

Variance and the Square Root of Time

Adding Independent Normal Changes

Suppose a Markov variable has a normally distributed one-year change with mean zero and variance one:

$$\Delta X_{1\text{ year}} \sim \mathcal{N}(0,1).$$

If annual changes are independent, the two-year change is the sum of two independent normal variables. Means add and variances add:

$$\Delta X_{2\text{ years}} \sim \mathcal{N}(0,2).$$

The standard deviation is therefore $\sqrt{2}$, not 2.

More generally, if the variance rate is one per year, then over a time interval of length T years,

$$\Delta X_T \sim \mathcal{N}(0,T), \qquad \mathrm{sd}(\Delta X_T) = \sqrt{T}.$$

> **Square-root-time rule.** Variance scales linearly with time. Standard deviation scales with the square root of time. This is why daily volatility is annual volatility divided by about $\sqrt{252}$, and annual volatility is daily volatility multiplied by about $\sqrt{252}$.

Example 1 (Scaling uncertainty)
If a process has annual standard deviation 20%*, then under the square-root rule its one-month standard deviation is approximately*

$$20\%\sqrt{\frac{1}{12}} = 5.77\%.$$

Its one-week standard deviation, using 52 weeks per year, is

$$20\%\sqrt{\frac{1}{52}} = 2.77\%.$$

The one-month uncertainty is not one twelfth of the annual uncertainty because standard deviations do not add across independent periods.

Wiener Processes

Definition

A Wiener process, also called standard Brownian motion, is the basic source of randomness in the continuous-time stock price model.

Definition 3 (Wiener process)
A process z_t is a Wiener process if:

1. *for a small interval Δt, the increment is*

$$\Delta z = \varepsilon\sqrt{\Delta t}, \qquad \varepsilon \sim \mathcal{N}(0,1);$$

2. *increments over non-overlapping time intervals are independent.*

Thus,

$$\mathbb{E}[\Delta z] = 0, \qquad \mathrm{Var}(\Delta z) = \Delta t, \qquad \mathrm{sd}(\Delta z) = \sqrt{\Delta t}.$$

Over a longer interval T,

$$z_T - z_0 \sim \mathcal{N}(0,T).$$

Why Brownian Paths Are Jagged

Brownian increments are proportional to $\sqrt{\Delta t}$. For small Δt, $\sqrt{\Delta t}$ is much larger than Δt. For example, if $\Delta t = 0.0001$, then $\sqrt{\Delta t} = 0.01$. The random term is therefore large relative to deterministic time drift over very short intervals.

This is the source of the strange-looking Brownian path. It is continuous, but it is nowhere differentiable in the ordinary sense. There is no well-defined instantaneous slope.

> **Intuition.** A smooth curve has small movements of order dt. Brownian motion has small movements of order $\sqrt{dt}$. As the time grid becomes finer, the path keeps revealing new short-run variation instead of settling down to a smooth tangent line.

Differential Notation

In stochastic calculus, we write

$$dz$$

for the infinitesimal increment of a Wiener process. This notation should be read through its discrete approximation:

$$dz \approx \varepsilon\sqrt{dt}.$$

The notation is compact, but the scale is important. The object dz is random and has variance dt.

Example 2 (Distribution of a Brownian value)
Suppose $z_0 = 25$. Since $z_T - z_0 \sim \mathcal{N}(0, T)$,

$$z_T \sim \mathcal{N}(25, T).$$

At $T = 1$, the standard deviation is 1. At $T = 5$, the standard deviation is $\sqrt{5} = 2.236$.

Generalized Wiener Processes

Adding Drift and Scale

The standard Wiener process has zero drift and variance rate one. A generalized Wiener process adds a deterministic drift and rescales the noise:

$$dx = a\,dt + b\,dz,$$

where a and b are constants.

Over a small time interval,

$$\Delta x = a\Delta t + b\varepsilon\sqrt{\Delta t}.$$

Therefore,

$$\mathbb{E}[\Delta x] = a\Delta t, \qquad \mathrm{Var}(\Delta x) = b^2\Delta t.$$

Over a longer interval T,

$$x_T - x_0 \sim \mathcal{N}(aT, b^2T).$$

Definition 4 (Drift rate and variance rate)
In $dx = a\,dt + b\,dz$, the drift rate is a and the variance rate is b^2. The diffusion coefficient b determines the standard deviation per square root of time.

Example 3 (Cash position)
A company's cash balance, measured in thousands, follows

$$dx = 20\,dt + 30\,dz.$$

The drift rate is 20 thousand per year and the variance rate is $30^2 = 900$. *If* $x_0 = 50$, *then after one year*

$$x_1 \sim \mathcal{N}(70, 900),$$

so the mean is 70 and the standard deviation is 30. After six months,

$$x_{0.5} \sim \mathcal{N}(60, 450),$$

so the standard deviation is $\sqrt{450} = 21.21$.

What Drift and Diffusion Do

The drift term $a\,dt$ moves the expected path. The diffusion term $b\,dz$ creates dispersion around that path. In a very short interval, the stochastic term is usually the visually dominant part because it is of order $\sqrt{dt}$. Over longer horizons, the drift accumulates linearly while uncertainty grows with the square root of time.

> **Useful mental picture.** Drift controls the direction of the cloud's center. Volatility controls the width of the cloud. A single realized path can move against the drift for a long time because the diffusion term is random.

ItoProcesses

State-Dependent Drift and Volatility

A generalized Wiener process has constant a and b. An Itoprocess allows them to depend on the current state and time:

$$dx = a(x,t)\,dt + b(x,t)\,dz.$$

Over a small interval,

$$\Delta x \approx a(x,t)\Delta t + b(x,t)\varepsilon\sqrt{\Delta t}.$$

This is a local approximation. During the short interval from t to $t + \Delta t$, we evaluate drift and volatility at the current state (x, t).

> **Local thinking.** Continuous-time finance is local. We ask what happens over the next instant given the current state, then aggregate the implications. This is why derivatives can be priced by local hedging arguments.

If $a(x,t)$ and $b(x,t)$ depend only on current x and t, the process is Markov. If they depend on the entire past path, the process is generally not Markov.

The Stock Price Process

Why Not Constant Dollar Volatility?

It is tempting to model a stock as

$$dS = a\,dt + b\,dz.$$

This says the expected dollar change and dollar volatility are constant. That is not natural for stocks. A 2 dollar daily standard deviation is huge for a 10 dollar stock and small for a 200 dollar stock.

Stock investors usually think in percentage returns. If required expected return is 10 percent per year, that means 10 percent whether the stock price is 20 or 200. Likewise, volatility is usually quoted as a percentage of price.

Geometric Brownian Motion

The standard continuous-time stock price model is

$$dS = \mu S\,dt + \sigma S\,dz,$$

or equivalently

$$\frac{dS}{S} = \mu\,dt + \sigma\,dz.$$

Definition 5 (Geometric Brownian motion)
A stock price follows geometric Brownian motion if

$$\frac{dS}{S} = \mu\,dt + \sigma\,dz,$$

where μ is the expected continuously compounded rate of return and σ is the volatility.

The proportional change over a short interval is approximately

$$\frac{\Delta S}{S} = \mu\Delta t + \sigma\varepsilon\sqrt{\Delta t}.$$

Thus,

$$\frac{\Delta S}{S} \sim \mathcal{N}(\mu\Delta t, \sigma^2\Delta t) \quad \text{approximately for small } \Delta t.$$

Interpretation of μ and σ. The parameter μ moves the expected return. The parameter σ is the annualized standard deviation of continuously compounded returns. In derivative pricing, σ is central. The real-world μ will disappear from the no-arbitrage price in Lecture 6.

Example 4 (Weekly stock return approximation)
A stock has expected return $\mu = 15\%$ and volatility $\sigma = 30\%$ per year. For one week, take $\Delta t = 1/52 = 0.01923$. Then

$$\frac{\Delta S}{S} = 0.15(0.01923) + 0.30\varepsilon\sqrt{0.01923} = 0.00288 + 0.0416\varepsilon.$$

The expected weekly return is about 0.288% *and the weekly standard deviation is about* 4.16%.

Why the Model Is Called Geometric

Without uncertainty,

$$dS = \mu S\, dt$$

has solution

$$S_T = S_0 e^{\mu T}.$$

The stock grows geometrically, not linearly. With uncertainty, the same multiplicative idea remains: shocks affect percentage changes, so prices tend to stay positive and compound over time.

Monte Carlo Simulation

Euler Simulation

The simplest simulation uses the discrete approximation

$$S_{t+\Delta t} = S_t + \mu S_t \Delta t + \sigma S_t \varepsilon \sqrt{\Delta t} = S_t \left(1 + \mu \Delta t + \sigma \varepsilon \sqrt{\Delta t}\right).$$

At each step we draw a new independent $\varepsilon \sim \mathcal{N}(0, 1)$.

Example 5 (One simulated step)
Let $S_t = 100$, $\mu = 0.15$, $\sigma = 0.30$, $\Delta t = 1/52$, and suppose the normal draw is $\varepsilon = 0.52$. Then

$$\Delta S = 100(0.00288 + 0.0416(0.52)) = 2.45.$$

The simulated next price is 102.45.

Exact Lognormal Simulation

For geometric Brownian motion with constant μ and σ, we can simulate more accurately using the exact transition:

$$S_{t+\Delta t} = S_t \exp\left[\left(\mu - \frac{1}{2}\sigma^2\right)\Delta t + \sigma\sqrt{\Delta t}\varepsilon\right].$$

This formula comes from applying Ito's lemma to $\ln S$, derived below.

Practical point. Euler simulation is easy and useful for intuition. Exact lognormal simulation is preferred for geometric Brownian motion because it preserves positivity and avoids time-step approximation error for this specific model.

Volatility and Parameters

Annualization

If σ is annual volatility, then the standard deviation of the continuously compounded return over T years is

$$\sigma\sqrt{T}.$$

For one trading day with 252 trading days in a year, it is

$$\frac{\sigma}{\sqrt{252}}.$$

Conversely, if daily log-return standard deviation is s_d, annual volatility is approximately

$$s_d\sqrt{252}.$$

Example 6 (Daily and annual volatility)
If annual volatility is 32%, *daily volatility is approximately*

$$\frac{0.32}{\sqrt{252}} = 0.0202,$$

or 2.02%. *If daily volatility is* 1.4%, *annual volatility is approximately*

$$0.014\sqrt{252} = 0.222,$$

or 22.2%.

Why μ Is Harder Than σ

Expected returns are hard to estimate because the signal is small relative to noise. Volatility is easier because squared returns reveal the scale of fluctuations even when average returns are imprecise.

For derivative pricing, this distinction matters. In the Black-Scholes-Merton model, the derivative price depends on volatility, not on the stock's real-world expected return. The reason is dynamic hedging: the stock's risk premium affects both the derivative and the hedge in a way that cancels out.

Correlated Wiener Processes

Two Sources of Risk

Many derivatives depend on more than one risk factor. Examples include options on exchange rates, spread options, quanto options, and derivatives with stochastic volatility or stochastic interest rates.

Suppose

$$dx_1 = a_1\,dt + b_1\,dz_1, \qquad dx_2 = a_2\,dt + b_2\,dz_2.$$

If the Brownian shocks have correlation ρ, then

$$\mathrm{Corr}(dz_1, dz_2) = \rho.$$

In differential notation this is often summarized by the multiplication rule

$$dz_1\,dz_2 = \rho\,dt.$$

Simulating Correlated Normals

To simulate two correlated standard normal shocks, draw independent $u, v \sim \mathcal{N}(0,1)$ and set

$$\varepsilon_1 = u, \qquad \varepsilon_2 = \rho u + \sqrt{1-\rho^2}\,v.$$

Then ε_1 and ε_2 each have standard normal distributions and their correlation is ρ.

Example 7 (Constructing correlated shocks)
Let $\rho = 0.60$. Suppose $u = 1.10$ and $v = -0.40$. Then

$$\varepsilon_1 = 1.10, \qquad \varepsilon_2 = 0.60(1.10) + \sqrt{1-0.60^2}(-0.40).$$

Since $\sqrt{1-0.36} = 0.80$,

$$\varepsilon_2 = 0.66 - 0.32 = 0.34.$$

The second shock shares a common component with the first shock but also has independent residual variation.

The Rules of Stochastic Calculus

Quadratic Variation

The defining scale of Brownian motion is

$$dz \sim \sqrt{dt}.$$

Therefore,

$$(dz)^2 \sim dt.$$

This is the key difference from ordinary calculus. Terms involving $(dt)^2$ and $dt\,dz$ are negligible, but $(dz)^2$ is not negligible because it is of order dt.

The informal multiplication table is

$$(dt)^2 = 0, \qquad dt\,dz = 0, \qquad (dz)^2 = dt.$$

For correlated Brownian motions,

$$dz_i\,dz_j = \rho_{ij}\,dt.$$

Why the table works. In a small interval, dt is tiny and dz is of size $\sqrt{dt}$. Thus $(dt)^2$ is much smaller than dt, and $dt\,dz$ is of size $dt^{3/2}$, also much smaller than dt. But $(dz)^2$ is of size dt and survives in the limit.

Taylor Expansion With a Random Term

Let x follow

$$dx = a(x,t)\,dt + b(x,t)\,dz.$$

Let $G(x,t)$ be a smooth function. The second-order Taylor expansion is

$$dG = G_x\,dx + G_t\,dt + \frac{1}{2}G_{xx}(dx)^2 + G_{xt}\,dx\,dt + \frac{1}{2}G_{tt}(dt)^2 + \cdots.$$

Using the stochastic multiplication table, the only second-order term that survives is $(dx)^2$:

$$(dx)^2 = (a\,dt + b\,dz)^2 = b^2(dz)^2 = b^2 dt.$$

Therefore,

$$dG = G_x(a\,dt + b\,dz) + G_t\,dt + \frac{1}{2}G_{xx}b^2 dt.$$

Ito's Lemma

Theorem 1 (Ito's lemma, one state variable)
If

$$dx = a(x,t)\,dt + b(x,t)\,dz,$$

and $G = G(x,t)$ is sufficiently smooth, then

$$dG = \left(G_x a + G_t + \frac{1}{2}G_{xx}b^2\right)dt + G_x b\,dz.$$

The new term relative to ordinary calculus is

$$\frac{1}{2}G_{xx}b^2\,dt.$$

It appears because the stochastic variable has nonzero quadratic variation.

> **Financial interpretation.** If G is an option price and x is the stock price, then G_x is delta and G_{xx} is gamma. Ito's lemma says that option value changes because of delta exposure, direct passage of time, and curvature interacting with variance.

Example: Squaring a Brownian Motion

Let $x = z$ be a Wiener process, so $dx = dz$. Take

$$G(x) = x^2.$$

Then $G_x = 2x$, $G_{xx} = 2$, and $G_t = 0$. Since $a = 0$ and $b = 1$, Ito's lemma gives

$$d(x^2) = \frac{1}{2}(2)(1)\,dt + 2x\,dz = dt + 2x\,dz.$$

Ordinary calculus would give only $2x\,dx$. Stochastic calculus adds dt. That extra term is the accumulated variance of Brownian motion.

Example: Exponential of Brownian Motion

Let

$$dx = \mu\,dt + \sigma\,dz, \qquad G = e^x.$$

Here $G_x = G$ and $G_{xx} = G$. Ito's lemma gives

$$dG = G\left(\mu + \frac{1}{2}\sigma^2\right)dt + G\sigma\,dz.$$

The exponential transformation changes the drift by $\frac{1}{2}\sigma^2$. Convexity plus randomness raises the expected change in the exponential.

Applying Ito's Lemma to Stock Prices

General Function of a Stock Price

If

$$dS = \mu S\,dt + \sigma S\,dz,$$

and $G = G(S, t)$, then $a = \mu S$ and $b = \sigma S$. Ito's lemma becomes

$$dG = \left(G_S\mu S + G_t + \frac{1}{2}G_{SS}\sigma^2 S^2\right)dt + G_S\sigma S\,dz.$$

The same dz appears in both dS and dG. This common source of uncertainty is what allows a derivative to be hedged locally using the underlying stock.

Bridge to Lecture 6. If an option and the underlying stock are driven by the same Brownian shock, a portfolio of the option and the stock can be chosen so that the dz terms cancel. The resulting locally riskless portfolio must earn the risk-free rate. That is the engine behind the Black-Scholes- Merton differential equation.

Application: Forward Price

For a non-dividend-paying stock with constant interest rate r, the forward price for maturity T at time t is

$$F(S,t) = Se^{r(T-t)}.$$

The derivatives are

$$F_S = e^{r(T-t)}, \qquad F_{SS} = 0, \qquad F_t = -rSe^{r(T-t)}.$$

Using Ito's lemma,

$$dF = \left(\mu Se^{r(T-t)} - rSe^{r(T-t)}\right) dt + \sigma Se^{r(T-t)} dz.$$

Since $F = Se^{r(T-t)}$,

$$dF = (\mu - r)F\,dt + \sigma F\,dz.$$

Interpretation. The forward price has the same volatility as the stock because it is proportional to the stock price. Its real-world drift is $\mu - r$ because the forward price already compounds the spot price forward at rate r.

The Lognormal Property

Applying Ito's Lemma to $\ln S$

Let

$$G(S) = \ln S.$$

Then

$$G_S = \frac{1}{S}, \qquad G_{SS} = -\frac{1}{S^2}, \qquad G_t = 0.$$

With $dS = \mu S\,dt + \sigma S\,dz$,

$$d\ln S = \left[\frac{1}{S}\mu S + \frac{1}{2}\left(-\frac{1}{S^2}\right)\sigma^2 S^2\right] dt + \frac{1}{S}\sigma S\,dz.$$

Therefore,

$$d \ln S = \left(\mu - \frac{1}{2}\sigma^2\right) dt + \sigma\, dz.$$

The Ito correction. The expected log return is not μ. It is $\mu - \frac{1}{2}\sigma^2$. Volatility lowers expected log growth because losses hurt compounded wealth more than equal-sized gains help it.

Distribution of Future Stock Prices

Integrating from 0 to T,

$$\ln S_T - \ln S_0 = \left(\mu - \frac{1}{2}\sigma^2\right) T + \sigma(z_T - z_0).$$

Since $z_T - z_0 \sim \mathcal{N}(0, T)$,

$$\ln S_T \sim \mathcal{N}\left(\ln S_0 + \left(\mu - \frac{1}{2}\sigma^2\right) T, \sigma^2 T\right).$$

Equivalently,

$$S_T = S_0 \exp\left[\left(\mu - \frac{1}{2}\sigma^2\right) T + \sigma\sqrt{T}\varepsilon\right], \qquad \varepsilon \sim \mathcal{N}(0, 1).$$

Definition 6 (Lognormal distribution)
A positive random variable is lognormally distributed if its natural logarithm is normally distributed. Under geometric Brownian motion, S_T is lognormal.

Mean, Median, and the Volatility Drag

The expected stock price under the real-world process is

$$\mathbb{E}[S_T] = S_0 e^{\mu T}.$$

The median stock price is

$$\text{median}(S_T) = S_0 e^{(\mu - \frac{1}{2}\sigma^2)T}.$$

The mean exceeds the median because the lognormal distribution is right-skewed: rare large upward outcomes pull up the average.

Example 8 (Distribution after one year)
Let $S_0 = 100$, $\mu = 12\%$, $\sigma = 30\%$, and $T = 1$. Then

$$\ln S_1 \sim \mathcal{N}\left(\ln 100 + (0.12 - 0.045), 0.09\right).$$

The mean log return is 7.5% and the standard deviation of the log return is 30%. The expected stock price is

$$100e^{0.12} = 112.75,$$

while the median stock price is

$$100e^{0.075} = 107.79.$$

From Real-World to Risk-Neutral Dynamics

Lecture 5 is mainly about stochastic calculus, but it is useful to preview the pricing move. Under the real-world probability measure, the stock model is

$$\frac{dS}{S} = \mu\, dt + \sigma\, dz.$$

Under the risk-neutral measure used for pricing a non-dividend-paying stock, the drift becomes the risk-free rate:

$$\frac{dS}{S} = r\, dt + \sigma\, dz^Q.$$

The volatility is unchanged in the basic Black-Scholes-Merton model.

> **Do not over-interpret risk-neutral probabilities.** Risk-neutral probabilities are pricing probabilities, not forecasts. They are constructed so that discounted asset prices are martingales and expected payoffs can be discounted at the risk-free rate.

The corresponding risk-neutral terminal distribution is

$$\ln S_T \sim \mathcal{N}\left(\ln S_0 + \left(r - \frac{1}{2}\sigma^2\right) T, \sigma^2 T\right).$$

This is the distribution used to derive the Black-Scholes-Merton formula.

Common Mistakes

1. **Adding standard deviations across time.** Variances add; standard deviations scale with the square root of time.
2. **Treating dz like dt.** Brownian increments are of order $\sqrt{dt}$, so $(dz)^2$ contributes a dt term.
3. **Forgetting the Itocorrection.** If $dS/S = \mu dt + \sigma dz$, then $d \ln S = (\mu - \sigma^2/2)dt + \sigma dz$.
4. **Confusing arithmetic and log returns.** The expected arithmetic instantaneous return is μ, but expected log growth is $\mu - \sigma^2/2$.
5. **Using real-world probabilities for pricing.** Option pricing uses risk-neutral dynamics after no-arbitrage arguments, not the stock's actual expected return.

Optional Extension: Fractional Brownian Motion

Standard Brownian motion has independent increments and variance proportional to elapsed time:

$$\mathrm{Var}(z_t - z_s) = t - s.$$

Fractional Brownian motion generalizes this to

$$\mathrm{Var}(X_t - X_s) = \sigma^2 (t-s)^{2H},$$

where H is the Hurst exponent. When $H = 1/2$, the usual Brownian scaling is recovered. When $H \neq 1/2$, increments are no longer independent in the same way. Such models are useful in modern rough-volatility work, but the classical Black-Scholes-Merton model uses ordinary Brownian motion.

Worked Problems

Example 9 (From annual to quarterly uncertainty)
A stock has volatility 24% *per year. The standard deviation of its continuously compounded return over one quarter is*

$$0.24\sqrt{0.25} = 0.12.$$

Thus the quarterly log-return standard deviation is 12%.

Example 10 (Probability of a log return event)
Let $S_0 = 50$, $\mu = 8\%$, $\sigma = 20\%$, *and* $T = 1$. *What is* $\mathbb{P}(S_T > 55)$ *under the real-world process?*

Since

$$\ln S_T \sim \mathcal{N}\left(\ln 50 + (0.08 - 0.5(0.20)^2), 0.20^2\right),$$

we have

$$\ln S_T \sim \mathcal{N}(\ln 50 + 0.06, 0.04).$$

Therefore

$$\mathbb{P}(S_T > 55) = \mathbb{P}\left(Z > \frac{\ln(55) - \ln(50) - 0.06}{0.20}\right).$$

The numerator is approximately $0.0953 - 0.06 = 0.0353$, *so the z-score is* 0.1765. *The probability is about* $1 - N(0.1765) = 0.43$.

Example 11 (Ito's lemma for a power payoff)
Let $G(S) = S^n$ *and*

$$dS = \mu S\, dt + \sigma S\, dz.$$

Then

$$G_S = nS^{n-1}, \qquad G_{SS} = n(n-1)S^{n-2}.$$

Ito's lemma gives

$$dG = \left[n\mu S^n + \frac{1}{2} n(n-1)\sigma^2 S^n \right] dt + n\sigma S^n dz.$$

Since $G = S^n$,

$$\frac{dG}{G} = \left[n\mu + \frac{1}{2} n(n-1)\sigma^2 \right] dt + n\sigma dz.$$

Convex power payoffs have drift terms that increase with variance.

Example 12 (Risk-neutral lognormal distribution)
A non-dividend-paying stock has $S_0 = 80$, $r = 5\%$, $\sigma = 25\%$, *and* $T = 0.5$. *Under the risk-neutral measure,*

$$\ln S_T \sim \mathcal{N}\left(\ln 80 + \left(0.05 - \frac{1}{2} 0.25^2 \right) 0.5, 0.25^2 (0.5) \right).$$

The log-price mean is

$$\ln 80 + (0.05 - 0.03125)0.5 = \ln 80 + 0.009375,$$

and the log-price standard deviation is

$$0.25\sqrt{0.5} = 0.1768.$$

Checklist for Students

After this lecture you should be able to:

- distinguish discrete-time, continuous-time, discrete-state, and continuous-state stochastic processes;
- state and interpret the Markov property;
- explain why variance, not standard deviation, adds over independent periods;
- define a Wiener process and describe its increment distribution;
- explain why Brownian paths are continuous but jagged;
- write and interpret a generalized Wiener process;
- write and interpret an Itoprocess;
- explain why stock prices are modeled with proportional drift and proportional volatility;
- simulate one step of geometric Brownian motion;
- convert volatility across time units using the square-root-time rule;

- construct correlated normal shocks from independent normal shocks;
- use the stochastic multiplication rules $(dt)^2 = 0$, $dt\,dz = 0$, and $(dz)^2 = dt$;
- state Ito's lemma for one stochastic variable;
- apply Ito's lemma to simple functions such as x^2, e^x, and $\ln S$;
- derive the process for $\ln S$ when S follows geometric Brownian motion;
- state the lognormal distribution of S_T;
- distinguish real-world and risk-neutral stock price dynamics.

Practice Questions

1. A process has annual variance rate 4. What are the variance and standard deviation of the change over 3 months?
2. If z_t is a Wiener process and $z_0 = 10$, what is the distribution of z_2?
3. Explain in words why Brownian motion does not have an ordinary derivative.
4. A generalized Wiener process follows $dx = 5dt + 2dz$. What are the mean, variance, and standard deviation of $x_T - x_0$ for $T = 3$?
5. A cash balance follows $dx = 12dt + 8dz$ and starts at 40. What is the distribution after one half year?
6. A stock follows $dS/S = 0.10dt + 0.25dz$. What are μ and σ? Interpret both.
7. For the stock in the previous question, compute the approximate mean and standard deviation of the proportional change over one month.
8. If annual volatility is 18%, what is daily volatility using 252 trading days?
9. If daily volatility is 1.8%, what is annualized volatility using 252 trading days?
10. Simulate one Euler step for $S = 100$, $\mu = 0.08$, $\sigma = 0.20$, $\Delta t = 1/12$, and $\varepsilon = -0.75$.

11. Draw independent $u = 0.30$ and $v = 1.20$. Construct two standard normal shocks with correlation $\rho = 0.50$ using the formula in the notes.

12. Explain why $(dz)^2$ is treated as dt but $dt\,dz$ is treated as zero.

13. State Ito's lemma for $dx = a(x,t)dt + b(x,t)dz$ and $G = G(x,t)$.

14. Use Ito's lemma to find the process for $G = x^3$ when $dx = a\,dt + b\,dz$ with constant a and b.

15. Use Ito's lemma to find the process for $G = 1/S$ when $dS = \mu S dt + \sigma S dz$.

16. Derive $d \ln S$ from $dS/S = \mu dt + \sigma dz$.

17. If $S_0 = 100$, $\mu = 10\%$, $\sigma = 40\%$, and $T = 1$, write the distribution of $\ln S_T$.

18. For the previous question, compute the expected stock price and the median stock price.

19. Under risk-neutral dynamics for a non-dividend-paying stock, what replaces μ? What happens to σ?

20. Explain why the real-world expected return μ is not the key input for Black-Scholes-Merton option pricing.

21. A derivative value is $G(S,t)$. Explain why the fact that S and G share the same Brownian shock is essential for delta hedging.

22. What part of the Lecture 5 material is a bridge to Lecture 6 rather than a pricing result by itself?